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Language statistics and individual differences in processing primary metaphors

Abstract: Research in cognitive linguistics has emphasized the role of embodiment in metaphor comprehension, with experimental research showing activation of perceptual simulations when processing metaphors. Recent research in conceptual processing has demonstrated that findings attributed to embodied cognition can be explained through language statistics. The current study investigates whether language statistics explain processing of primary metaphors and whether this effect is modified by the gender of the participant. Participants saw word pairs with valence (Experiment 1: *good–bad*), authority (Experiment 2: *doctor–patient*), temperature (Experiment 3: *hot–cold*), or gender (Experiment 4: *male–female*) connotations. The pairs were presented in either a vertical configuration (*X* above *Y* or *Y* above *X*) matching the primary metaphors (e.g., HAPPY IS UP, CONTROL IS UP) or a horizontal configuration (*X* left of *Y* or *Y* left of *X*) not matching the primary metaphors. Even though previous research has argued that primary metaphor processing can best be explained by an embodied cognition account, results demonstrate that statistical linguistic frequencies also explain the response times of the stimulus pairs both in vertical and horizontal configurations, because language has encoded embodied relations. In addition, the effect of the statistical linguistic frequencies was modified by participant gender, with female participants being more sensitive to statistical linguistic context than male participants.

Keywords: perceptual simulation, embodied cognition, language statistics, statistical linguistic frequencies, symbolic cognition, symbol interdependency, conceptual metaphor, cognitive task, gender differences

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1 Introduction

Studies in cognitive linguistics demonstrate that conceptual metaphors emphasize relations between concepts and physical properties. These metaphors are thought to be understood through their relationships to physical space (Gibbs 1994, 2006; Kövecses 2005; Lakoff and Johnson 1980; 1999). For instance, expressions like *I am feeling up* or *I am flying high* mark the primary metaphor HAPPY IS UP, with positive terms being associated with 'up' properties and negative terms with 'down' properties. Similarly, the primary metaphor CONTROL IS UP associates concepts related to power and authority to high vertical positions. For example, a *supervisor* might find that in his *lofty* position, he is on *top* of a situation with everything *under* control. Accordingly, being subordinate is associated with low vertical positions; a new employee is on the *bottom* of the totem pole and must look up to his superiors. The source of these spatial properties is embodied: "These spatial orientations arise from the fact that we have bodies of the sort we have and that they function as they do in our physical environment" (Lakoff and Johnson 1980: 14). That is, "human conceptual processing is deeply grounded in embodied metaphor" (Gibbs 2006: 122).

Theories of embodied cognition suggest that for any concept to become meaningful it must be grounded in the perceptual world around us and its comprehension requires activation of perceptual simulations (Barsalou 1999; Bergen 2012; Glenberg 1997; Zwaan 2004). Many researchers in cognitive linguistics support this embodied view, suggesting that a variety of metaphors are understood through perceptual simulations of the body within physical space (Gibbs 1994, 2006; Lakoff and Johnson 1980; 1999). For instance, Wilson and Gibbs (2007) found that when subjects performed an action congruent with a metaphorical statement, they were faster to comprehend the statements than when statements were incongruent with an action. After performing or imagining a grasping motion, a subject would be faster to process the metaphorical phrase *grasp a concept* than when performing or imagining other incongruent motions (Wilson and Gibbs 2007). Such findings demonstrate that metaphor comprehension can be explained by bodily action. Systematic linguistic analyses show that numerous additional metaphors are also grounded in bodily mechanisms (e.g., hunger, pressure, temperature, space, and emotion) (Gibbs et al. 2004; Kövecses 1986; Nayak and Gibbs 1990). For instance, abstract metaphors such as those linked to time seem to be processed through embodied mechanisms related to space. For example, the past is likely to be thought of as being behind us, as evidenced by phrases like *back in the day* or *that day is behind us now,* whereas the future is thought of as being ahead of us with phases like *are you looking forward to it*? (Gibbs 2006; Lakoff and Johnson 1999). In a number of studies, Boroditsky and colleagues demonstrated that time is indeed grounded in space with subjects being likely to organize time spatially, such that spatial representations and manipulations influence how subjects process and conceptualize time (Boroditsky 2000, 2001; Boroditsky and Ramscar 2002). For example, subjects moving forward in space are likely to think of time in terms of the TIME PASSING IS A MOVING OBSERVER metaphor (as opposed to the TIME PASSING IS A MOVING OBJECT metaphor) because they themselves are physically moving forward in space. In addition, subjects reading sentences describing motion in a particular direction are faster to process such sentences after witnessing motion in that same direction than after witnessing motion in a different direction (Dils and Boroditsky 2010).

The association between valence or authority concepts and specific physical locations can also be readily explained by embodied cognition. For example, evidence for the primary metaphor HAPPY IS UP comes from experiments that show that when an image or word representing a positive concept is presented on the top of a computer screen, comprehension is facilitated (Crawford et al. 2006; Meier and Robinson 2005). For the primary metaphor CONTROL IS UP similar findings have been obtained. Studies have shown that when an authority word (e.g., *master*) was presented on the upper part of the screen, response times (RTs) were faster and recall was better than when a non-authority word (e.g., *servant*) was presented in that same position (Meier and Robinson 2004; Meier et al. 2007; Schnall and Clore 2004; Schubert 2005). In other words, comprehending primary metaphors such as HAPPY IS UP and CONTROL IS UP seems to be facilitated through the perceptual simulation of upward/downward directions, supporting the idea that primary metaphors are understood through their relationship to physical space.

In fact, results similar to the aforementioned primary metaphor experiments have been obtained using concepts with literal upward and downward locations, such as flying and swimming animals presented in the upper or the lower part of the screen (Pecher et al. 2010; Šetić and Domijan 2007). Similarly, when word pairs with a perceptual order such as *attic* and *basement* are presented to participants, one above the other, iconic presentations (e.g., *attic* above *basement*) are processed faster than reverse-iconic presentations (e.g., *basement* above *attic*) (Zwaan and Yaxley 2003). Such iconicity findings have also been obtained with other object words (Estes et al. 2008) and motion verbs (Meteyard et al. 2007). Over the last decade findings like these have accumulated, lending support to the claim that linguistic symbols are grounded in modality specific perceptual and motor systems (de Vega et al. 2008; Pecher and Zwaan 2005; Semin and Smith 2008). However, although conceptual metaphors are grounded in perception and action, perceptual simulations of the external world are not requisite for understanding and thinking about these metaphors (Boroditsky and Ramscar 2002).

A number of theories have cautioned against a unilateral embodied account of cognition. Paivio (1986) has extensively shown that both the verbal channel and the non-verbal channel play important roles during cognition. Barsalou et al. (2008) and Louwerse (2007, 2008, 2011) have also argued that perceptual simulations and symbolic relationships work together to represent conceptual knowledge. Louwerse (2007, 2011) proposed the Symbol Interdependency Hypothesis, arguing that language encodes perceptual relations. For instance, language has encoded up and down relations by typically placing the *up* concept before the *down* concept, as is the case in many binomials (e.g., it is more common to say *up and down*, *head to toe*, *top and bottom* than *down and up*, *toe and head*, and *bottom and top*) (Cooper and Ross 1975). That is, perceptual cues are encoded linguistically, such that language users can rely on the linguistic system as a shortcut to the perceptual system. Consequently, comprehension can be explained both by a statistical linguistic approach and a perceptual simulation approach. For instance, Louwerse (2008) demonstrated that findings of iconicity attributed to perceptual simulations, such as the facilitative iconicity effect when the word *attic* is placed above *basement*, can also be explained by statistical linguistic frequencies (the word order *attic–basement* is more frequent than the order *basement–attic*). Louwerse and Jeuniaux (2010) also demonstrated that both perceptual simulation and linguistic context contributed to conceptual processing, but the relative importance of the two factors was modified by the stimulus (words or pictures) and cognitive task (semantic judgment or iconicity task). Louwerse and Connell (2011) showed that findings initially attributed to perceptual simulation, such as increased processing times for modality shifts, can in fact also be attributed to statistical linguistic frequencies. Also, faster processing was best explained by the linguistic system, and slower processing was best explained by the perceptual system. These findings demonstrate that individuals rely on perceptual and linguistic information to varying degrees, based on a number of factors.

The current paper had two goals, (1) to determine if, in addition to embodied mechanisms, language statistics can also account for metaphor processing and (2) to explore how individual differences impact conceptual processing. First, researchers have made the claim that processing metaphors is so fundamentally embodied that no other factors are expected to be relevant for comprehension. Although it may seem obvious that frequency-based effects might be expected, some argue that metaphor comprehension is due solely to perceptual simulation (Glenberg 1997; Van Dantzig et al. 2008). Contrary to this claim that metaphor comprehension is embodied only, Boroditsky and Ramscar (2002) and Gibbs (2006) have stated that embodiment effects do not account for the entirety of metaphor comprehension, but other factors are likely relevant too. One such candidate is language itself. Following the Symbol Interdependency Hypothesis, we predicted that if language encodes perceptual information, statistical linguistic frequencies will also be able to explain conceptual processing (Louwerse 2008; Louwerse and Jeuniaux 2010). In several prior experiments, findings could be attributed to perceptual simulation, and also to statistical linguistic frequencies. However, all of these studies use stimuli found in the physical world, such as objects (Louwerse 2008; Louwerse and Jeuniaux 2010), or modalities (Louwerse and Connell 2011). Whether the same applies to non-literal relations, such as metaphors, is an open question. In the following experiments we explore the possibility that metaphorical and literal thought are processed similarly enough, such that statistical linguistic frequencies are not only important during literal language comprehension but also when processing primary metaphors. We predicted that findings that have been predominantly attributed to perceptual simulations, such as the faster processing when authority words are positioned above non-authority words compared to the reverse, can also be attributed to statistical linguistic frequencies. Therefore, although embodied simulations account for a significant portion of metaphor comprehension, perhaps other factors also play an important role (Boroditsky and Ramscar 2002; Gibbs 2006; Louwerse, 2007, 2008, 2011). If language statistics can explain conceptual processing of word pairs in their vertical configuration it would provide evidence that primary metaphors can be explained by language statistics and embodiment together. If language statistics also explains processing of word pairs in their horizontal configuration – a configuration that provides little support for an embodiment account – it will provide further evidence for the importance of language statistics during conceptual processing (Louwerse 2008).

Second, with regards to individual differences, the question can be raised to what extent different participants are more or less accustomed to relying upon statistical linguistic frequencies when processing conceptual metaphors. Studies have shown that the relative prominence of statistical linguistic frequencies or perceptual simulation is modulated by the stage of conceptual processing (Louwerse and Connell 2011), the cognitive task (Louwerse and Jeuniaux 2010), and the stimulus (Louwerse and Jeuniaux 2010). Yet it is unclear whether individual differences also modulate the relative importance of linguistic context. For instance, those participants with enhanced language skills may show an inclination to process information in a statistical linguistic fashion. This possibility can be explored through gender differences. Males tend to show greater general spatial ability and females show greater general and spoken language ability (Bourke and Adams 2011; Kimura 2000; Kramer et al. 1997; Linn and Peterson 1985). A large body of research finds gender differences between language and between spatial ability. For example, males often outperform females on spatial tasks (Benbow and Stanley 1983; Casey et al. 2001). Similarly, females are likely to have superior language skills for a variety of tasks such as verbal fluency tasks, semantic categorization tasks, and verbal memory tasks (Andreano and Cahill 2009; Burman et al. 2008; Bornstein et al. 2000), as well as language advantages throughout development (Wei et al. 2012). In addition, males are more likely to develop language disorders (Liederman et al. 2005; Rutter et al. 2004). Based on the aforementioned tendencies in gender, we hypothesized that because females may have a greater affinity to encode information linguistically, this affinity would generalize to a semantic judgment task such that linguistic factors would be hypothesized to better predict female RTs than male RTs.

In four experiments we aimed to answer these two questions (i.e., that of the role of statistical linguistic context and the effect of gender on statistical linguistic frequencies). In each experiment male and female participants responded to word pairs with either valence (positive and negative), authority (superior and inferior), temperature (hot and cold), or gender (male and female) connotations presented in either a vertical or a horizontal configuration.

2 Experiment 1: Valence

In Experiment 1, we presented participants with word pairs that were opposites on a valence dimension (e.g., *happy–sad*) either in a vertical configuration (*happy* above/below *sad*) or horizontal configuration (*happy* left/right of *sad*). We investigated the extent to which statistical linguistic frequencies explained RTs to the iconic and reverse-iconic presentation of primary metaphor word pairs. In addition, we investigated whether participant gender interacted with the language statistics.

2.1 Method

2.1.1 Participants

Seventy-four native English speaking undergraduates at the University of Memphis (53 females) participated for extra credit in a Psychology course. Thirty-four subjects (24 females) were randomly assigned to the vertical presentation condition and forty subjects (29 females) were randomly assigned to the horizontal presentation condition.

2.1.2 Materials

The experiment consisted of 50 pairs of words that were opposites on a valence dimension (e.g., *happy–sad*). To avoid a response bias towards the iconicity of the word pairs, 100 filler items consisted of word pairs without a positive-negative relation (e.g., *blade–walnut*), with half of the pairs having a high semantic association and half having a low semantic association as determined by latent semantic analysis (LSA), a computational linguistic technique that measures the similarity in meaning between word pairs, but ignores an order relation (Landauer et al. 2007) (high semantic association: $cos = .44$; low semantic relation: $cos = .18$).

2.1.3 Procedure

Participants were asked to judge the semantic relatedness of word pairs presented on an 800 × 600 computer screen running E-Prime software (Psychology Software Tools Inc., Pittsburgh, PA). Words were presented one above another in the vertical condition, and next to one another in the horizontal condition. Upon the presentation of a word pair, participants indicated whether the pair was related in meaning by pressing designated yes or no keys. All word pairs were randomly ordered for each participant to negate any order effects. To ensure participants understood the task, participants completed five practice trials before beginning the experimental task.

2.2 Results and discussion

RTs above 2.5 *SD* from the mean per condition, per subject, were removed from the analysis, affecting 4.17% of the RT data.

2.2.1 Statistical linguistic frequencies

As in previous studies (Louwerse 2008; Louwerse and Connell 2011; Louwerse and Jeuniaux 2010) we operationalized statistical linguistic frequencies as the log frequency of a–b (e.g., *happy–sad*) or b–a (e.g., *sad–happy*) order of word pairs. The order frequency of all word pairs within 3–5 word grams was obtained using the large Web 1T 5-gram corpus (Brants and Franz 2006).

Statistical linguistic frequencies indeed confirmed the HAPPY IS UP metaphor, with positive-negative ordered word pairs having a higher frequency than negative-positive word pairs, $t(44)$, 10.81, $p < .001$, $M = 12.15$, $SD = 2.77$ vs. $M = 10.47$, $SD = 2.66$.

For both vertical and horizontal conditions, a mixed-effect regression analysis was conducted on RTs with the linguistic frequency as a fixed factor and participants and items as random factors (Baayen et al. 2008). The model was fitted using the restricted maximum likelihood estimation (REML) for the continuous variable (RT). F-test denominator degrees of freedom were estimated using the Kenward-Roger's degrees of freedom adjustment to reduce the chances of Type I error (Littell et al. 2002).

For the vertical configuration of word pairs, statistical linguistic frequencies explained RTs, $F(1, 84.55) = 58.80$, $p < .001$, $R^2 = .41$ with higher frequencies yielding lower RTs. For the horizontal condition, the statistical linguistic frequencies again explained RTs, $F(1, 86.21) = 82.44$, $p < .001$, $R^2 = .49$, again with higher frequencies yielding lower RTs. These findings show that the HAPPY IS UP metaphor is encoded in language and that statistical linguistic frequencies explain the processing of this primary metaphor, both in vertical and horizontal configurations of the word pairs.

2.2.2 Participant gender effects

Having demonstrated that statistical linguistic context explained RTs, we next addressed the question whether the effect is modulated by participant gender. To account for the differences in number of males (n = 21) versus females (n = 53), Markov-chain Monte Carlo (MCMC) sampling generated 100 datasets with equal numbers of males and females. From this, we conducted 100 mixed effects analyses on RT for each gender with the linguistic variable as our fixed factor. We then compared these t-values by gender. For the vertical condition, a t-test revealed that the t-values for each gender significantly differed, $t(198) = -4.80$, $p < .001$, with men having lower t-values than women (as linguistic frequency was more likely to predict female RTs than male RTs). However, for the horizontal condition, the t-values again were not significantly different between genders, $t(198) = -0.84$, $p = .40$ (Figure 1).

These results show the RT effects in both horizontal and vertical configuration of word pairs can be explained by statistical linguistic frequencies that encode the primary metaphor HAPPY IS UP. The effects are modulated by individual differences, with male participants relying less on the linguistic frequencies than female participants in the vertical condition, but not in the horizontal condition, where the vertical nature of such metaphors is not emphasized perceptually. Whether these findings can be extended to other metaphors is investigated in the next three experiments.

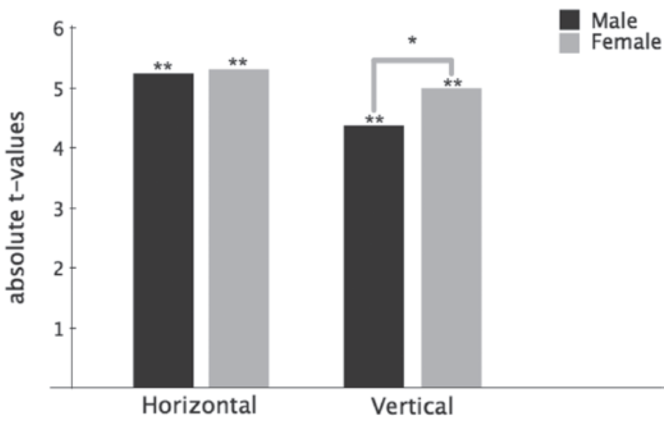

Fig. 1: Male and female t-values from mixed effects analyses on RT from MCMC generated values in vertical and horizontal conditions of valence word pairs. Larger t-values indicate greater reliance on the linguistic factor. (* denotes significant effects at p < .05, ** denotes significant RT effects at p < .001)

3 Experiment 2: Authority

Experiment 2 was identical to Experiment 1 except that we investigated whether statistical linguistic context determined processing of word pairs related to the Lakoff and Johnson (1999) CONTROL IS UP metaphor.

3.1 Method

3.1.1 Participants

Seventy-nine different native English speaking undergraduate at the University of Memphis (55 females) participated for extra credit in a Psychology course. Forty-one subjects (28 females) were randomly assigned to the vertical condition and 38 subjects (27 females) were randomly assigned to the horizontal condition.

3.1.2 Materials

The experiment consisted of 30 pairs of words that were opposites on an authoritative dimension (e.g., *parent–child*) and 60 filler items without an authority relation, with half of the pairs having a high semantic association and half having a low semantic association as determined by LSA (high semantic association: *cos* = .45; low semantic relation: *cos* = .20).

3.1.3 Procedure

The procedure was identical to Experiment 1.

3.2 Results and discussion

Subjects whose RTs fell more than 2.5 *SD* from the mean per condition, per subject were removed from the analysis, affecting 8.64% of the data.

3.2.1 Statistical linguistic frequencies

Statistical linguistic frequencies were operationalized in the same way as for Experiment 1. Again, evidence was found that the primary metaphor was encoded in language, with powerful–powerless word pairs being more frequent than powerless–powerful word pairs, $t(21) = 6.42$, $p < .001$, $M = 11.36$, $SD = 2.71$ vs. $M = 9.70$, $SD = 3.01$.

For both the vertical and horizontal configurations, mixed-effect regressions were again conducted on RTs with linguistic frequency as the fixed factor and participants and items as random factors.

As in Experiment 1, for the vertical condition linguistic frequencies explained the RTs, $F(1, 45.77) = 32.71$, $p < .001$, $R^2 = .42$, with higher frequencies yielding lower RTs. For the horizontal condition, the linguistic factor was also related to the RT, $F(1, 45.07) = 16.25$, $p < .001$, $R^2 = .26$, with higher frequencies yielding lower RTs. These findings again show that primary metaphors are encoded in language, and that both the vertical and horizontal configuration of conceptual metaphors can be explained by the linguistic system, confirming previous findings that statistical linguistic frequencies explain conceptual processing.

3.2.2 Participant gender effects

MCMC sampling generated 100 datasets with equal numbers of males and females. For the vertical configuration, a t-test revealed that the t-values for each gender significantly differed, $t(198) = 23.54$, $p < .001$, with women having higher t-values than men (as linguistic frequency was more likely to predict female

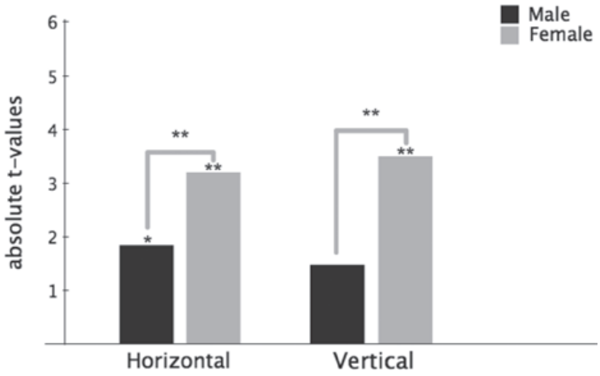

Fig. 2: Male and female t-values from mixed effects analyses on RT from MCMC generated values in vertical and horizontal conditions of authority word pairs. Larger t-values indicate greater reliance on the linguistic factor. (* denotes significant RT effects at $p < .05$, ** denotes significant RT effects at $p < .001$)

RTs than male RTs). The same was found for horizontal condition, $t(198) = 12.41$, $p < .001$ (Figure 2). The findings of Experiment 2 with authority words extended those in Experiment 1 with valence words. The linguistic factor again explained RTs both in the vertical and horizontal conditions, with effect sizes similar to those found in Experiment 1. Furthermore, as expected, female participants relied more on the statistical linguistic factor than male participants in both vertical and horizontal orientations.

4 Experiment 3: Temperature

Lakoff and Johnson (1980) claim that heat is affiliated with an upward location, whereas cold is affiliated with a downward location by giving the example *If you're too hot, turn the heat down*. Similarly, IJzerman and Semin (2009, 2010) give the example that *holding warm feelings toward someone* or *giving someone the cold shoulder* can be associated with positive (up) and negative (down) social relations, respectively. Furthermore, examples like *she was boiling with anger* suggest that an increase in body temperature is physiologically linked to anger and also the associated physical upwards motion (Pacini and Barnard 2011). In Experiment 3 we tested whether the finding for valence and authority can be extended to temperature. That is, we investigated whether hot and cold concepts can also be explained by statistical linguistic context, and whether these effects are modulated by participant gender.

4.1 Method

4.1.1 Participants

Seventy-two different undergraduate native English speakers at the University of Memphis (53 females) participated for extra credit in a Psychology course. Forty-one subjects (29 females) were randomly assigned to the vertical condition and 31 subjects (24 females) were randomly assigned to the horizontal condition.

4.1.2 Materials

The experiment consisted of 22 pairs of words that were opposites on an temperature dimension (e.g., *hot–cold*) with 44 filler items without a temperature relation, with half of the pairs having a high semantic association and half having a low semantic association (high semantic association: *cos* = .49; low semantic relation: *cos* = .20).

4.1.3 Procedure

The procedure was identical to that of Experiments 1 and 2.

4.2 Results and discussion

Subjects whose RTs fell more than 2.5 *SD* from the mean per condition, per subject were removed from the analysis, affecting 4.30% of the data.

4.2.1 Statistical linguistic frequencies

The operationalization of statistical linguistic frequencies was the same as for Experiments 1 and 2. Following the findings in the previous experiments that primary metaphors are encoded in language, we found that hot–cold word pairs were significantly more frequent than cold–hot word pairs, $t(17) = 6.12$, $p < .001$, $M = 11.55$, $SD = 3.40$ vs. $M = 10.67$, $SD = 3.37$.

For both conditions, the same mixed-effect analysis was conducted on RTs. As in Experiment 1 and 2, for the vertical condition the linguistic factor explained the RTs, $F(1, 33.16) = 31.31$, $p < .001$, $R^2 = .49$, with higher frequencies yielding lower RTs. For the horizontal condition, the linguistic factor also predicted RTs, $F(1, 33.51) = 30.63$, $p < .001$, $R^2 = .48$, with higher frequencies yielding lower RTs. These findings are similar to those obtained in the vertical condition, demonstrating that participants relied on statistical frequencies when processing word pairs related to temperature. Specifically, those word pairs matching the primary metaphor were processed faster than their counterparts, again both in vertical and horizontal configurations.

4.2.2 Participant gender effects

As in Experiment 1 and 2, MCMC sampling generated 100 datasets to account for differences in number of males and females. A t-test revealed that the t-values for each gender significantly differed in the vertical condition, $t(198) = 11.14$, $p < .001$, with women having higher t-values than men (as linguistic frequency was more likely to predict female RTs than male RTs). For the horizontal condition, similar results were obtained, $t(198) = 7.23$, $p < .001$ (Figure 3). The findings of Experiment 3 with temperature words replicated those in Experiment 1 and 2. The linguistic factor explained RTs for both configuration conditions, with effect sizes similar to those found in Experiments 1 and 2. Again, females relied more on the statistical linguistic factor than males in both conditions.

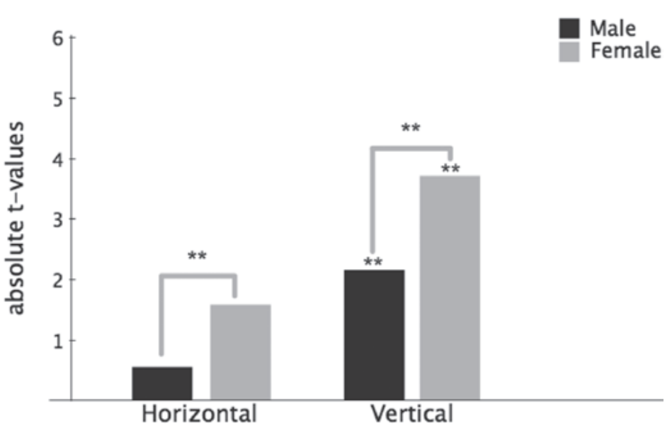

Fig. 3: Male and female t-values from mixed effects analyses on RT from MCMC generated values in vertical and horizontal conditions of temperature word pairs. Larger t-values indicate greater reliance on the linguistic factor. (** denotes significant RT effects at p < .001)

5 Experiment 4: Gender

Meier and Dionne (2009) extended the CONTROL IS UP metaphor to gender, whereby males are (stereotypically) considered to be powerful and females are (stereotypically) considered to be powerless. The prediction would then be that male concepts placed in upward physical location are processed faster than female concepts in that location. Meier and Dionne (2009) indeed demonstrated that pictures of female faces were thought to be more attractive when presented at the bottom of a screen (a location–concept match), whereas male faces were thought to be more attractive when presented at the top of a screen. Experiment 4 investigated word pairs such as *male–female* and *female–male* in vertical and horizontal configurations, similar to Experiments 1–3 investigating valence, authority, and temperature.

5.1 Method

5.1.1 Participants

Seventy-seven different undergraduate native English speakers at the University of Memphis (54 females) participated for extra credit in a Psychology course. Thirty-nine subjects (27 females) were randomly assigned to the vertical condition and 38 subjects (27 females) were randomly assigned to the horizontal condition.

5.1.2. Materials

The experiment consisted of 32 pairs of words that were opposites on a gender dimension (e.g., *male–female*). Sixty-four filler pairs lacked a gender relation, with half of the pairs having a high semantic association and half having a low semantic, (high semantic association: *cos* = .48; low semantic relation: *cos* = .22).

5.1.3 Procedure

Procedure was identical to that of Experiments 1–3.

5.2 Results and discussion

RTs that fell more than 2.5 *SD* from the mean per condition, per subject were removed from the analysis, affecting 2.45% of the data.

5.2.1 Statistical linguistic frequencies

Linguistic frequencies were operationalized in the same way as Experiments 1–3. Male–female pairs had higher frequencies than female–male pairs, $t(31) = 7.42$, $p < .001$, $M = 10.72$, $SD = 4.13$ vs. $M = 9.46$, $SD = 3.86$.

For both conditions, the same mixed-effect analysis was conducted on RTs as before. For the vertical condition statistical linguistic frequencies explained RTs, $F(1, 60.64) = 35.43$, $p < .001$, $R^2 = .37$, with higher frequencies yielding lower RTs. Linguistic frequencies also predicted RTs in the horizontal configuration, $F(1, 59.37) = 46.55$, $p < .001$, $R^2 = .44$, with higher frequencies yielding lower RTs. These findings are similar to those obtained in the other experiments and support the conclusion that processing of gender concepts in vertical and horizontal configurations can be explained by statistical linguistic frequencies.

5.2.2 Participant gender effects

As before, we tested if linguistic factors better predicted RTs for females. For the vertical condition, a t-test revealed that the t-values for each gender significantly differed in the vertical condition, $t(198) = 26.21$, $p < .001$, with women having higher t-values than men. For the horizontal condition, similar results were obtained, $t(198) = 30.51$, $p < .001$ (Figure 4). The findings of Experiment 4 with gender words replicate those in the previous Experiments. The linguistic factor explained RTs for both configuration conditions, with effect sizes similar to those found in Experiments 1 and 2. Furthermore, females relied more on the statistical linguistic factor than males in both conditions.

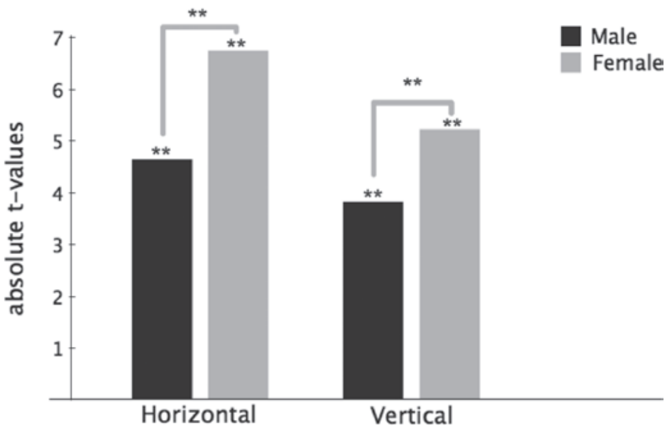

Fig. 4: Male and female t-values from mixed effects analyses on RT from MCMC generated values in vertical and horizontal conditions of gender word pairs. Larger t-values indicate greater reliance on the linguistic factor. (** denotes significant RT effects at p < .001)

6 General discussion

Many studies have made the argument that metaphorical thought is fundamentally embodied. For instance, experimental studies have been arguing this by showing how spatial location and valence are associated. In four experiments we investigated whether valence (e.g., *good–bad*), authority (e.g., *doctor–patient*), temperature (e.g., *hot–cold*), and gender (e.g., *male–female*) word pairs with matched iconicity were processed faster than mismatching word pairs. Finding evidence that metaphor comprehension is embodied is non-trivial, but begets the very important question whether alternative explanations can be dismissed (Van Dantzig et al. 2008), and whether there are alternative predictors (Louwerse 2008; Louwerse and Jeuniaux 2010). Here we investigated to what extent statistical linguistic frequencies explained RTs of word pairs related to valence (Experiment 1), authority (Experiment 2), temperature (Experiment 3), or gender (Experiment 4). The findings were not only considered in terms of significance, but also in terms of effect size. For example, Meier and Robinson (2004) obtained significant results, with positive words being processed faster on the top of the screen, and with negative words being processed faster on the bottom of the screen. Although this effect clearly supports an account of embodied cognition, the authors do not address the effect sizes associated with their results (R^2 = .16 for Experiment 1, and R^2 = .23 for Experiment 2). The effect sizes for the statistical linguistic frequencies reported in the current paper were consistently at least two times larger than effect sizes obtained from previous embodied cognition research such as Meier and Robinson (2004). Finally, different than most embodied cognition studies we analyzed RTs using a mixed effects model with both participant and items as random factors. Variance attributed to different participants and, importantly, variance attributed to different items was removed from the analysis to avoid the language-as-fixed fallacy effect, thereby yielding more reliable results (Clark 1973; Brysbaert 2007).

Do the results reported here suggest that statistical linguistic frequencies should be seen as an explanation for conceptual processing in lieu of perceptual simulation? Certainly not. We do not deny that embodiment offers some explanatory power during metaphor comprehension, however we only argue that it is not the only important factor at play. There are several reasons embodiment should not be overlooked. Other studies have found "subtle but pervasive" effects for the relation between vertical position and affect (Meier and Robinson 2004), temperature (IJzerman and Semin 2009, 2010), authority (Shubert 2005), and gender (Meier and Dionne 2009). In fact, some metaphors like these (e.g., HAPPY IS UP) seem to be near-universal, and this makes sense if metaphor comprehension is based on bodily perception and action. At the same time, many conceptual metaphors at the specific level vary cross-culturally and even regionally, suggesting that embodiment cannot entirely account for metaphor comprehension on its own (Kövecses 2005). In point of fact, cultures even vary with how they represent spatial information with relationship to their bodies, raising the question of how language comprehension can rely only on the body, when different cultures think of their bodies as being related to space in different ways. Kövecses (2005) argues that the body does not exist in isolation, instead it is molded by our environment, our history, and our interests thus introducing variance in how language is understood through the body. We argue that in addition to these factors, language itself influences metaphor comprehension, with statistical linguistic frequencies explaining RTs in the current experiments. But these findings beg the question of why it is the case that *happy–sad* (*hot–cold*, *teacher–student*, *male–female*) is more frequent than the reverse word pair orders? The answer is that language encodes perceptual relations. That is, when speakers formulate utterances, prelinguistic conceptual knowledge is translated into linguistic conceptualizations, so that as a function of language use, perceptual relations become encoded in language (Louwerse 2008). As shown in the current study, antonyms of valence, temperature, authority, and gender demonstrate that an embodied cognition account is not detached from a linguistic account. On the contrary, a statistical linguistic account complements an embodied cognition account (Louwerse 2011). The linguistic system has evolved to encode perceptual relations and these linguistic cues are employed by language users during their cognitive processes, as demonstrated in both vertical and horizontal configurations of word pairs.

In addition to investigating the role of statistical linguistic frequencies on conceptual processing, the current study investigated to what extent male and female participants were affected by these two factors differently. In prior studies we have investigated whether stimulus (word or picture), cognitive task (semantic judgment or iconicity task), or stage of processing (faster or slower RTs) affected the role of statistical linguistic frequencies and perceptual simulation differently. In the current study we investigated whether participant gender could be added to the series of modulators of statistical linguistic frequencies and conceptual processes. The findings in this study demonstrate that gender should indeed be considered a modulator. These experiments demonstrated that female participants typically relied more on statistical linguistic patterns. However, it should be noted that although we speculate that these findings might be due to differences in linguistic and spatial ability between genders we did not directly measure such constructs. Such a step would be useful in future research to ensure such effects are indeed driven by linguistic and spatial ability and are not simply due to other factors linked to gender.

The findings of the experiments reported in this study show that even though perceptual simulations contribute to RT differences, it is important to consider the extent to which other factors, such as language statistics, explain processing. It is also important to investigate to what extent such results can be generalized across types of stimulus, speed of processing, cognitive tasks, or – as demonstrated in the current paper – participant gender. The mind is embodied but is also linguistic, while participant gender tends to determine how much.

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